

US Bank Credit Stress Test

Panel econometrics | Model validation | Capital projection

End-to-end macro stress testing on public US bank data. This report documents the satellite NPL model, validation results, and stress test projections under the 2025 Federal Reserve severely adverse scenario.

RMSE (in-sample)	RMSE (OOS)	R-squared	Validation
0.31 pp	0.17 pp	0.972	12P / 4W

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1. At a glance

Item	Detail
Institutions	10 major US banks, 2005 Q1 to 2024 Q4
Credit metric	NPL ratio = $NCLNLS / LNLSNET \times 100$
Satellite model	Panel OLS, bank FE, HC1 robust SEs
Regressors	npl_lag1, real_gdp_growth_lag2, unemployment, mortgage_rate_lag1
In-sample backtest	RMSE 0.31 pp, MAE 0.18 pp
Out-of-sample	RMSE 0.17 pp, MAE 0.16 pp (2023-2024 holdout)
Validation	12 PASS, 4 WARN, 0 FAIL
Stress scenario	2025 Fed severely adverse, 9 quarters
Stress outcome	NPL roughly doubles, no 5% capital breach

2. What this project demonstrates

Skill area	Evidence in this project
Stress testing architecture	Decoupled scenario, satellite model, capital projection
Panel econometrics	Bank FE, AR persistence, macro transmission lags
Model development	Iterative lag selection to parsimonious 4-variable spec
Model validation	validate_model() with PASS/WARN/FAIL flags
Out-of-sample testing	2023-2024 holdout with bias reporting
Time-series diagnostics	Phillips-Perron and Zivot-Andrews unit root tests
Credit risk translation	NPL to losses via LGD, PPNR, equity roll-forward

3. Supervisory framework

Stage	Supervisory concept	Implementation
1	Macro scenario	Fed 2025 severely adverse CSV
2	Satellite model	Panel NPL regression on macro variables
3	Model validation	Backtest, OOS, diagnostics, unit roots
4	Balance-sheet projection	NPL to losses to PPNR to equity
5	Outcomes analysis	Capital troughs, loss totals, breach flags

4. Data construction

FDIC bank sample

CERT	Bank
628	JPMorgan Chase Bank NA
3510	Bank of America NA
3511	Wells Fargo Bank NA
7213	Citibank NA
6548	U.S. Bank NA
6384	PNC Bank NA
9846	Truist Bank
5649	Fifth Third Bank NA
6672	KeyBank NA
18409	Regions Bank

NPL ratio = $(\text{NCLNLS} / \text{LNLSNET}) \times 100$ from FDIC quarterly call reports.

FRED macro variables

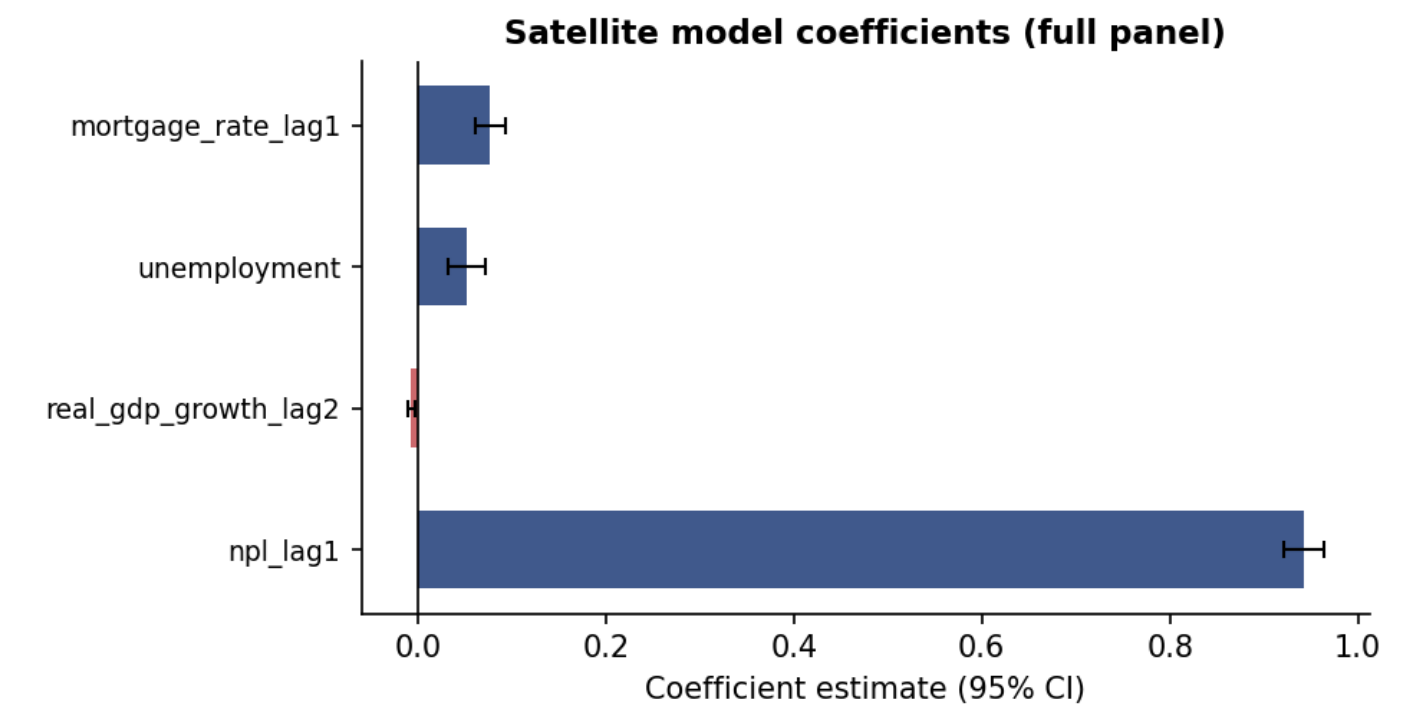
Variable	Series	Description
real_gdp_growth	A191RL1Q225SBEA	Real GDP growth q/q SAAR (%)
unemployment	UNRATE	Unemployment rate (%)
mortgage_rate	MORTGAGE30US	30-year mortgage rate (%)
t3m	TB3MS	3-month T-bill (%)
cpi_inflation	CPIAUCSL	CPI inflation YoY (%)
house_price_index	CSUSHPINSA	Case-Shiller HPI

Panel: 10 banks x 80 quarters = 800 rows. 780 observations after two-quarter lags.

5. Satellite model

$\text{NPL}(i,t) = a_i + \rho * \text{NPL}(i,t-1) + b_1 * \text{GDP}(t-2) + b_2 * \text{UR}(t) + b_3 * \text{Mtg}(t-1) + e(i,t)$. Pooled OLS with bank dummies and HC1 robust standard errors.

Figure 1. Estimated coefficients with 95% confidence intervals



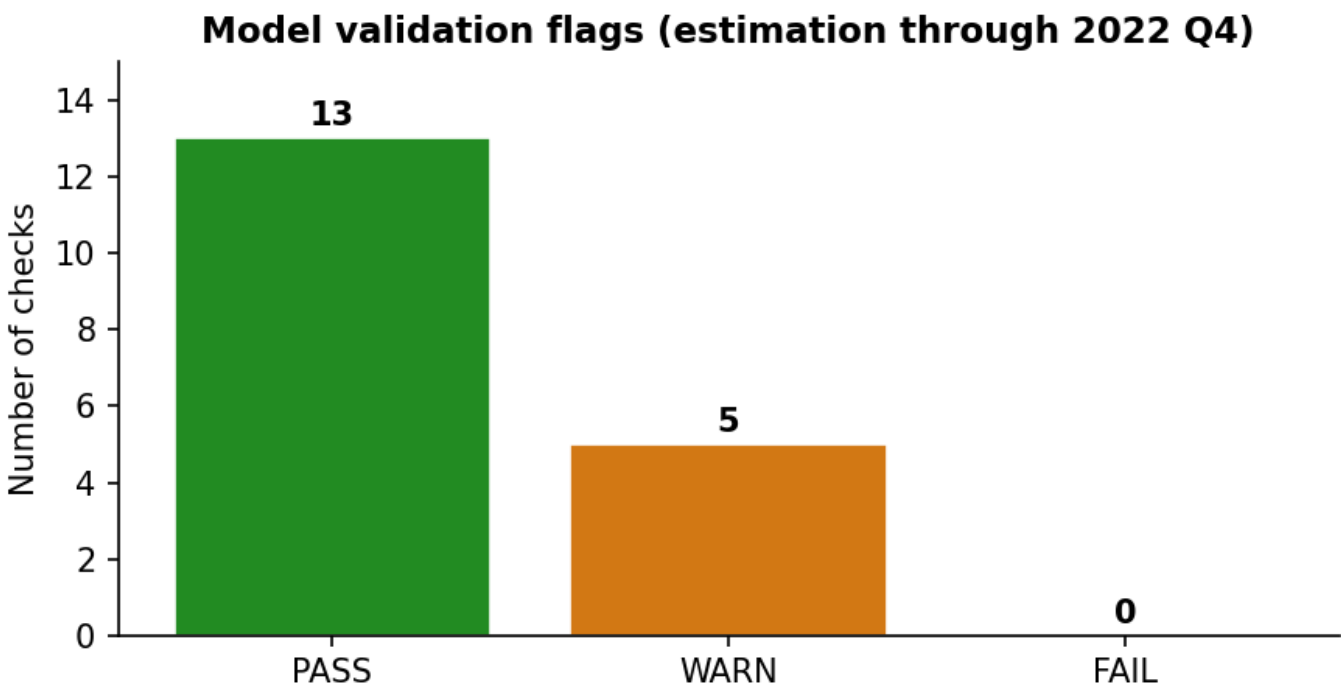
R-squared = 0.972. All coefficients significant at 1%. Signs match credit-cycle theory.

Specification search

Step	Action	Outcome
1	Baseline with 2 lags each	Over-parameterized for 10 banks
2	GDP lag 2 vs lag 1	Lag 2 better fit, theory-consistent sign
3	Unemployment contemporaneous vs lag	Contemporaneous more stable in forecast
4	Mortgage rate vs T-bill	Mortgage captured housing channel
5	Drop HPI, CPI, t3m	Parsimonious 4-variable final spec

6. Model validation

Figure 2. Automated validation check summary



Check	Category	Status	Detail
sign npl lag1	economic_plausibility	PASS	coef=0.9396, expected +
significance npl lag1	statistical	PASS	p-value=0.0000
sign real gdp growth lag2	economic_plausibility	PASS	coef=-0.0065, expected -
significance real gdp growth lag2	statistical	PASS	p-value=0.0002
sign unemployment	economic_plausibility	PASS	coef=0.0502, expected +
significance unemployment	statistical	PASS	p-value=0.0000
sign mortgage rate lag1	economic_plausibility	PASS	coef=0.0962, expected +
significance mortgage rate lag1	statistical	PASS	p-value=0.0000
r squared	statistical	PASS	R ² =0.9715
durbin watson	serial_correlation	WARN	DW=1.1182
ljung box	serial_correlation	WARN	residual autocorrelation detected
residual normality	distributional	WARN	Jarque-Bera p=0.0000
heteroskedasticity	distributional	PASS	Breusch-Pagan p=0.0000, HC1 robust SEs used in e
multicollinearity	statistical	WARN	high VIF (>10): unemployment
in sample rmse	backtesting	PASS	RMSE=0.3204 pp, MAE=0.1915 pp
gfc period rmse	backtesting	WARN	2008-2009 RMSE=0.7491 pp
out of sample rmse	backtesting	PASS	OOS RMSE=0.1726 pp (0.54x in-sample)
out of sample bias	backtesting	PASS	OOS bias=0.1445 pp

In-sample backtest

Metric	Value
RMSE	0.31 pp
MAE	0.18 pp

Figure 3. In-sample backtest: actual vs predicted NPL

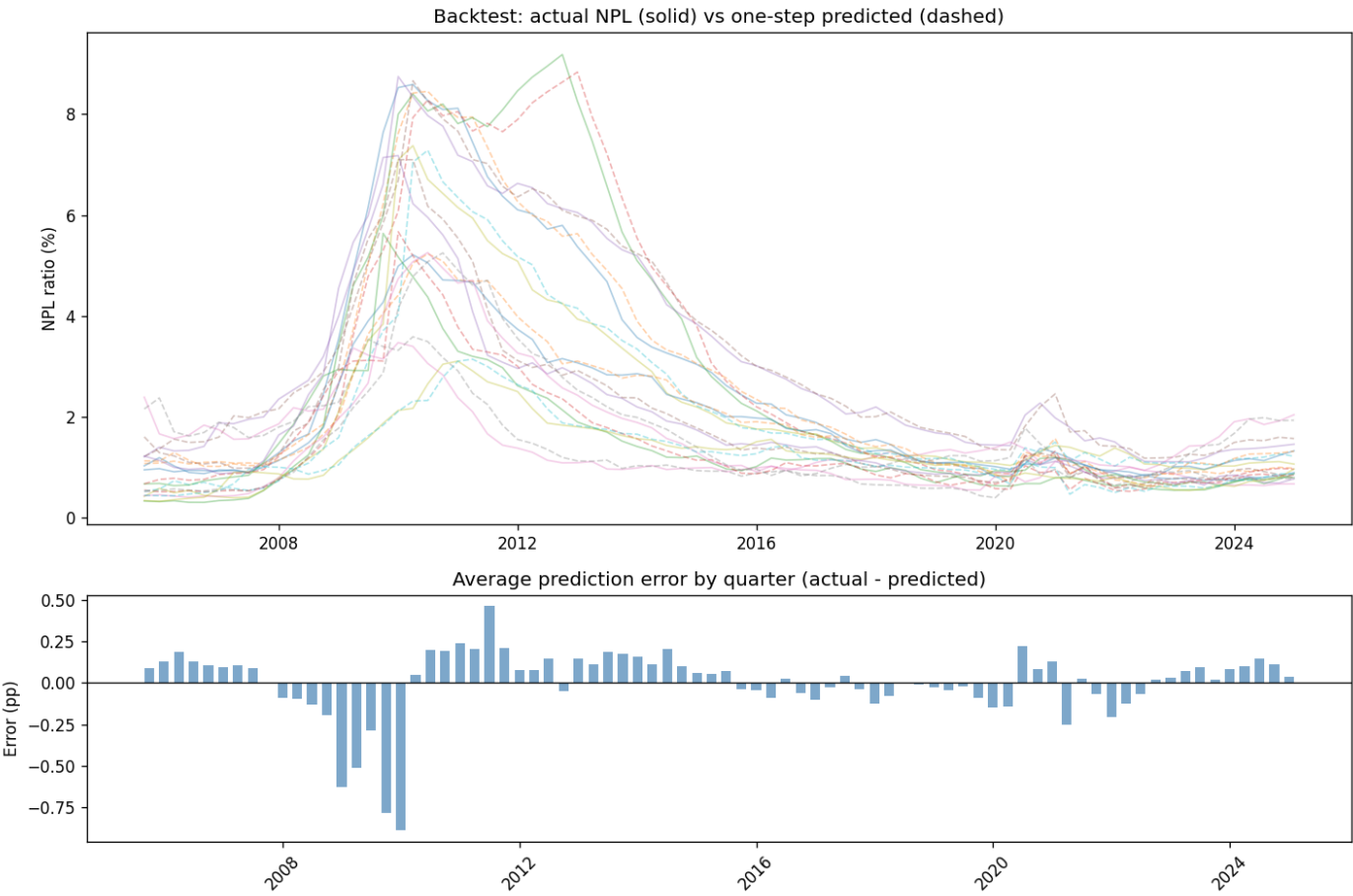
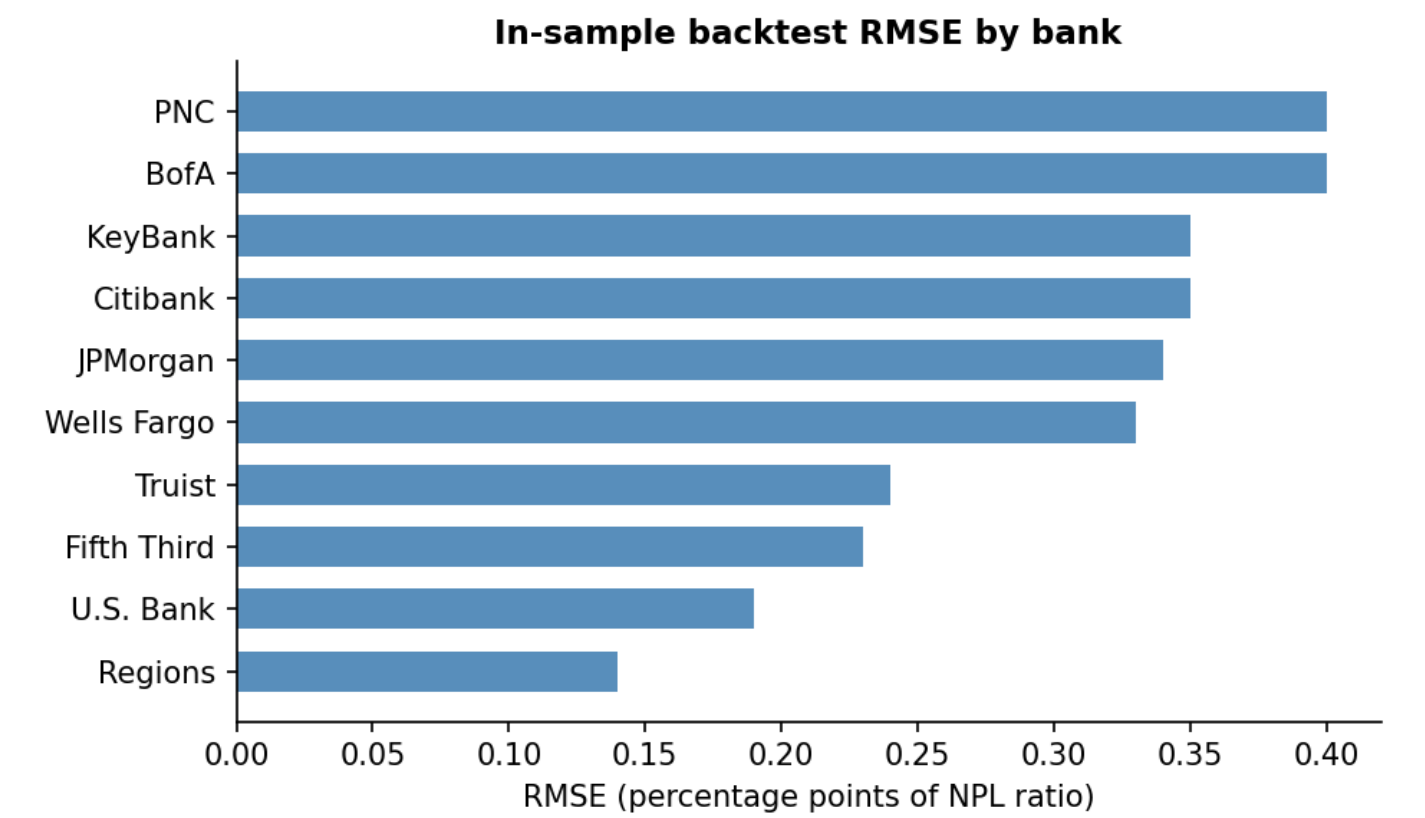


Figure 4. Backtest RMSE by bank

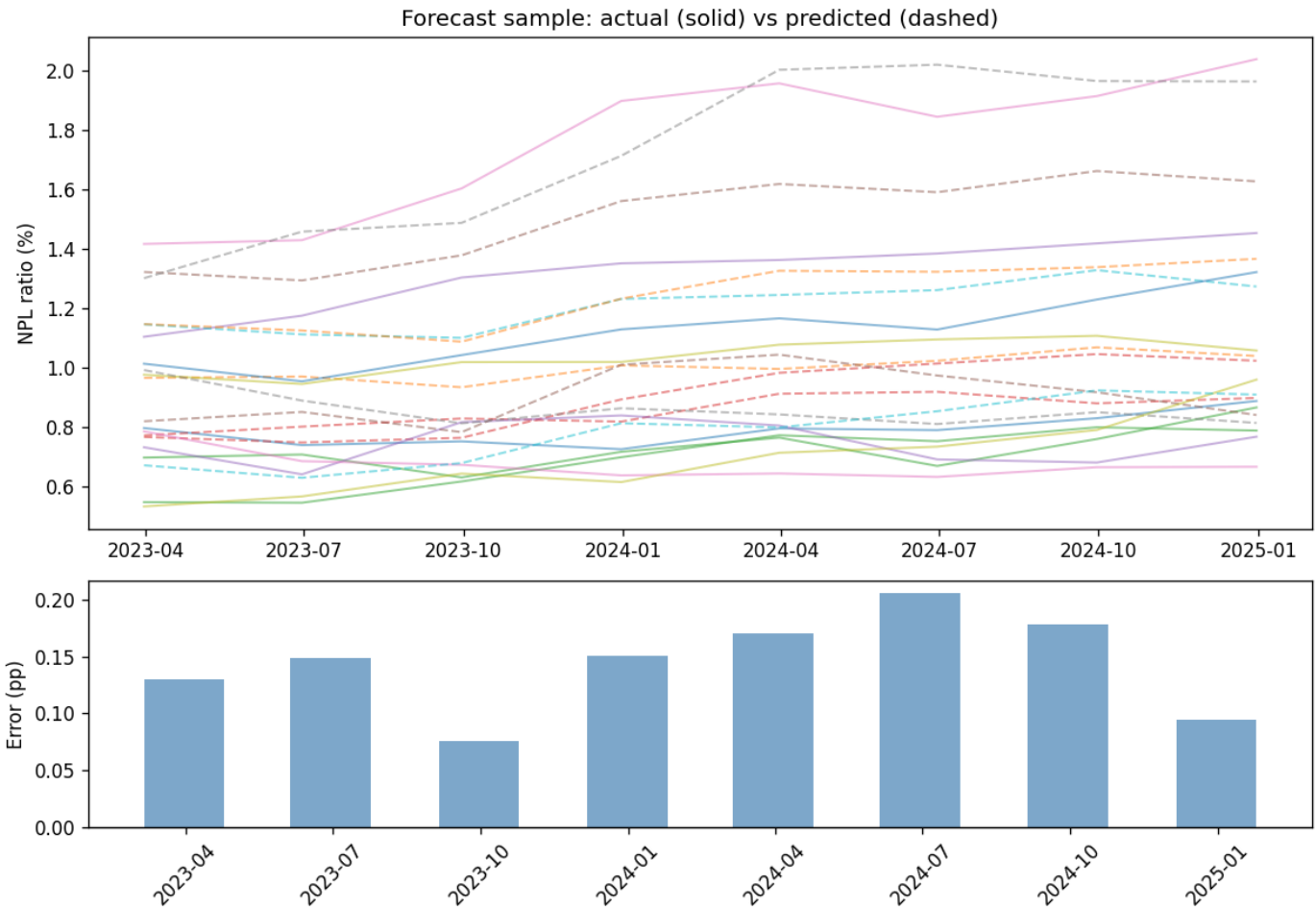


Out-of-sample forecast (2023 Q1 to 2024 Q4)

Metric	Value
RMSE	0.17 pp
MAE	0.16 pp
Bias	+0.14 pp (under-predicts)

Quarter	Actual (%)	Predicted (%)	Error (pp)
2023 Q1	0.86	0.99	+0.13
2023 Q2	0.84	0.99	+0.15
2023 Q3	0.91	0.98	+0.08
2023 Q4	0.96	1.11	+0.15
2024 Q1	1.01	1.18	+0.17
2024 Q2	0.97	1.18	+0.21
2024 Q3	1.02	1.20	+0.18
2024 Q4	1.08	1.17	+0.09

Figure 5. Out-of-sample forecast sample



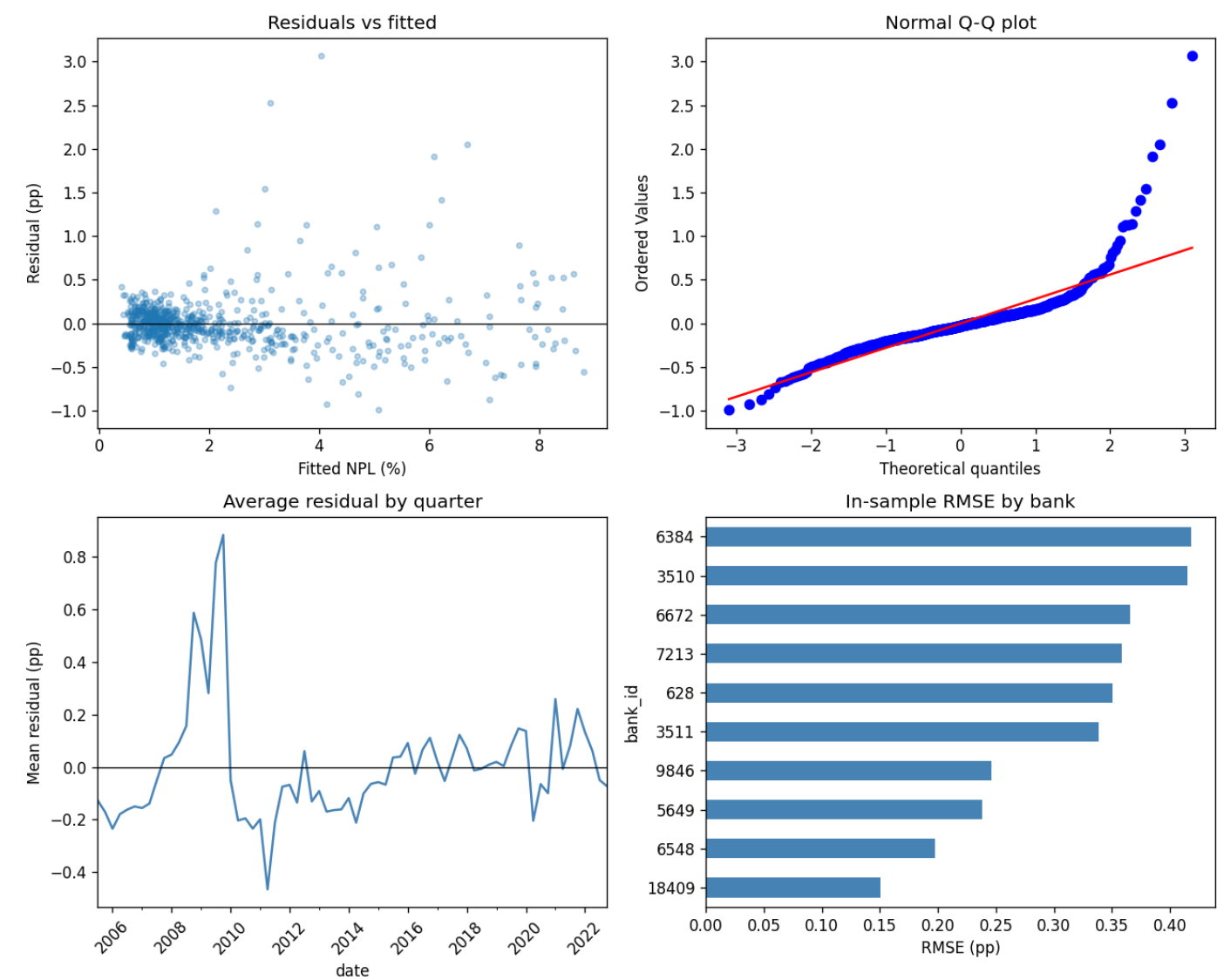
7. Regression diagnostics

Test	Result	Implication
Durbin-Watson	1.12	Positive residual autocorrelation
Ljung-Box	$p \sim 0$	Serial correlation in residuals
Jarque-Bera	$p \sim 0$	Fat tails from GFC period
Breusch-Pagan	$p \sim 0$	Heteroskedasticity, HC1 SEs used

Variable	VIF
npl_lag1	3.9
real_gdp_growth_lag2	1.1
unemployment	10.7
mortgage_rate_lag1	6.4

Period	RMSE (pp)	Comment
2005-2007	0.20	Pre-crisis calm
2008-2009	0.75	GFC nonlinearity not captured
2010-2014	0.29	Post-crisis repair
2015-2019	0.11	Low-volatility cycle
2020-2022	0.18	COVID partially captured

Figure 6. Residual diagnostic plots



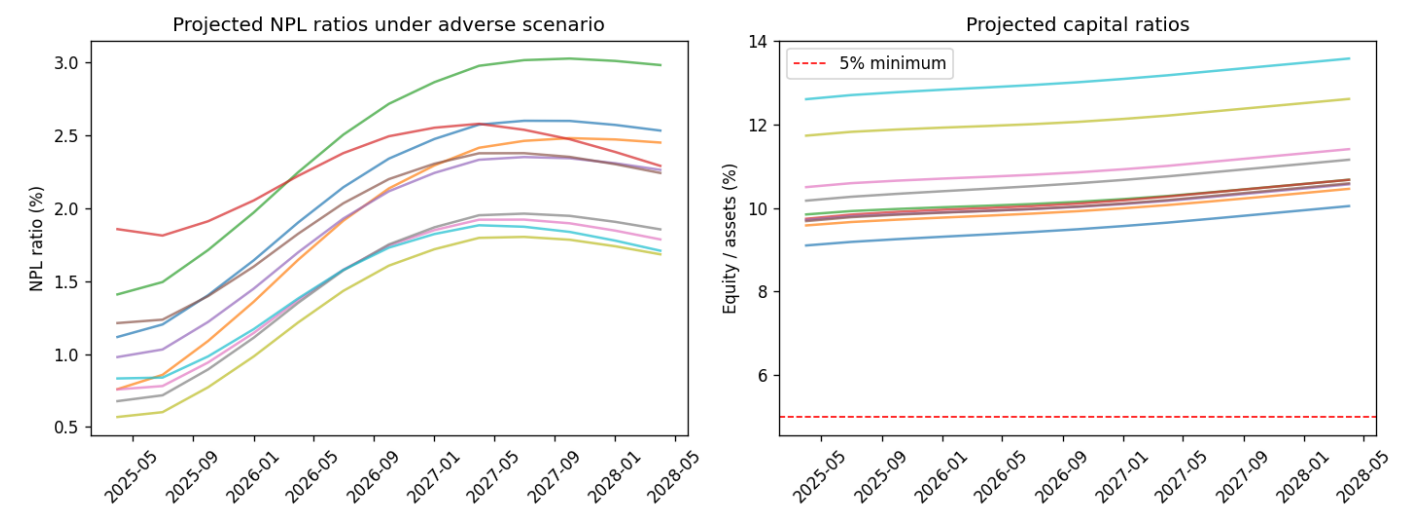
Unit root tests

Series	PP stationary?	ZA break	ZA stationary?
BofA NPL	No	~2007 Q4	No
JPMorgan NPL	No	~2007 Q4	Yes
real_gdp_growth	Yes	~2007 Q4	Yes
unemployment	No	~2007 Q4	No

8. Stress test results

2025 Fed severely adverse scenario from 2024 Q4 starting positions. Nine-quarter projection horizon.

Figure 7. Projected NPL and capital ratios under adverse scenario



Bank	Trough cap.	End cap.	Credit loss	Breach
JPMorgan Chase	9.10%	10.04%	\$10.3B	No
Bank of America	9.58%	10.46%	\$8.4B	No
U.S. Bank	9.68%	10.59%	\$2.0B	No
PNC	9.71%	10.56%	\$1.9B	No
Fifth Third	9.74%	10.67%	\$0.4B	No
Wells Fargo	9.84%	10.67%	\$6.3B	No
Citibank	10.17%	11.15%	\$3.7B	No
KeyBank	10.50%	11.41%	\$0.6B	No
Truist	11.73%	12.61%	\$1.7B	No
Regions	12.60%	13.57%	\$0.8B	No

Parameter	Stress	Default	Description
lgd	0.45	0.45	Loss given default on new NPLs
ppnr_roa_q	0.10%	0.30%	Quarterly PPNR under stress
min_ratio	5%	5%	Equity/assets breach threshold

9. Limitations and disclaimer

- Capital ratio is equity/assets, not risk-weighted CET1.
- PPNR is a flat ROA ratio. A second satellite for NII is the natural extension.
- No AFS/HTM mark-to-market under the scenario yield curve.
- Linear model does not capture GFC-style crisis nonlinearities.
- 10-bank sample is for methodology demonstration only.
- Residual serial correlation ($DW = 1.12$) warrants Newey-West robustness checks.
- Unemployment $VIF = 10.7$ requires rolling-window stability monitoring.

Educational project on public data. Not supervisory software. Outputs are not assessments of any real bank's solvency or capital adequacy.

10. Author

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GitHub: github.com/muhammadumarshinwari/us-bank-credit-stress-test